

**Analytical Solutions for Longwave Fluxes in Radiative–Convective
Equilibrium with Water Vapor**

Yi Zhang^a

^a *Courant Institute of Mathematical Sciences, New York University, New York, New York*

Corresponding author: Yi Zhang, y.zhang@nyu.edu

6 ABSTRACT: Water vapor is a major greenhouse gas in the Earth's atmosphere and its concentra-
7 tion is largely controlled by temperature according to the Clausius-Clapeyron relation. Building
8 on previous theoretical work, we show that, in an idealized radiative-convective equilibrium where
9 temperature decreases with height following adiabatic lapse rates and relative humidity is constant
10 in the vertical, the Planck function and optical depth of water vapor are approximately related by
11 a power law. The exponent γ of this power law is given by the ratio of the energy of a terres-
12 trial infrared photon ($h\nu$) to the latent heat release associated with the condensation of a single
13 water-vapor molecule (l_v), i.e., $\gamma = h\nu/l_v$. This power law originates from the Boltzmann factor
14 appearing in both the Clausius-Clapeyron relation and Planck's function. The power law enables
15 analytic solutions of the Schwarzschild equation for upward and downward longwave radiation
16 fluxes, from which atmospheric cooling and outgoing longwave radiation (OLR) follow directly as
17 functions of surface Planck emission and column-integrated optical depth. The nondimensional
18 parameter γ encapsulates the dual role of water vapor as both a condensable and a greenhouse gas,
19 and the same power law potentially applies to radiative-convective equilibrium in other planetary
20 atmospheres.

1. Introduction

Water vapor plays two roles in the atmosphere of the Earth (e.g. Pierrehumbert 2010). First, it is a condensable gas. With the oceans providing abundant surface reservoir, the concentration of water vapor is strongly limited by the capacity of the air to hold water vapor. This capacity is determined by the Clausius-Clapeyron relation between the saturation vapor pressure and temperature. Second, water vapor is a greenhouse gas. The bent structure of the H_2O molecule provides a permanent electric dipole moment that could interact with the electromagnetic field as the molecule rotates and vibrates. Transitions between rotational and rotation-vibrational states are accompanied by the absorption and emission of infrared photons, which coincide with the spectral range of much of Earth's thermal infrared emission.

These two roles of water vapor are coupled through the energy budget of the atmosphere. Earth's surface absorbs incoming solar radiation far more efficiently than the atmosphere above and, as a result, the surface is systematically warmer than the overlying air. This temperature discontinuity drives vigorous vertical mixing, preventing the troposphere from attaining radiative equilibrium (Manabe and Wetherald 1967). The troposphere thus persistently loses energy through infrared radiation (e.g. Trenberth et al. 2009) and this radiative cooling is dominated by water vapor (e.g. Fig. 4.29 in Wallace and Hobbs (2006)). Cooling of the tropospheric water vapor over the broad clear-sky regions is primarily balanced by condensation heating of water vapor in the convective updrafts that occupies a tiny fraction of area, resulting in the radiative-convective equilibrium (RCE) (Manabe and Wetherald 1967).

This work concerns the radiative transfer of water-vapor-dominated wavenumbers in an RCE. A main goal is to seek an analytic expression between the Planck emission $B_{\tilde{\nu}}(T)$ and the optical depth τ in the RCE. The motivation is that the Schwarzschild's equation for radiative transfer can be expressed in the optical depth space as follows

$$dI_{\tilde{\nu}} = \left(\pi B_{\tilde{\nu}}(T) - I_{\tilde{\nu}} \right) d\tau, \quad (1)$$

where $I_{\tilde{\nu}}$ is the spectral radiation flux of wavenumber $\tilde{\nu}$ with the unit $\text{W m}^{-2} \text{ cm}$. $I_{\tilde{\nu}}$ is intensified by the emission from the medium itself (first term on the right-hand side) and attenuated by the medium (second term on the right-hand side) of an incremental optical depth $d\tau$. The equation

48 would be analytically solvable if $B_{\tilde{\nu}}(T)$ can be expressed in τ . Thus, obtaining an explicit mapping
49 between the Planck function and optical depth enables analytic solutions for radiative fluxes.

50 Seeking a relationship between $B_{\tilde{\nu}}$ and τ is not entirely straightforward because Planck emission
51 (at a given $\tilde{\nu}$) only depends explicitly on temperature. Previous approaches have therefore proceeded
52 in three steps: first simplifying the relationship between $B_{\tilde{\nu}}$ and temperature T , then relating
53 temperature T to pressure p , and lastly relating p to τ . This sequence of approximations leads to
54 an effective power-law relationship between $B_{\tilde{\nu}}$ and τ (Jeevanjee and Fueglistaler 2020a).

55 The first step is usually done by considering a gray gas model, i.e., an atmosphere with spectrally
56 uniform absorptivity. In this case, the Planck emission $B_{\tilde{\nu}}(T)$ can be replaced by its spectral
57 integral, that is the Stefan-Boltzmann law σT^4 (e.g. Pierrehumbert 2010; Robinson and Catling
58 2012). The limitation of the gray gas model is that it neglects spectral structure, which can
59 be essential for radiative transfer and climate interactions (e.g., Jeevanjee and Fueglistaler 2020b;
60 Romps et al. 2022; Match et al. 2025; Koll and Cronin 2018). Alternatively, the spectral dependence
61 can be retained by approximating $B_{\tilde{\nu}}(T)$ as a power law in temperature (Jeevanjee and Fueglistaler
62 2020a).

63 The second step corresponds to specifying the temperature profile in pressure coordinates, which
64 is adiabatic in radiative–convective equilibrium (RCE). For a dry adiabatic troposphere, the lapse
65 rate is constant, implying a power-law relationship between T and p . A moist adiabatic troposphere
66 does not admit a closed-form $T(p)$ relation and approximations are introduced to retain analytical
67 tractability. One approach is to represent the smaller lapse rate of a moist adiabat but neglect its
68 change with height. Then T is approximated by a power law in p with a different exponent than its
69 dry counterpart (Robinson and Catling 2012; Jeevanjee and Fueglistaler 2020a).

70 The third step is most convenient for well-mixed gases such as CO_2 , for which the vertically
71 integrated path is proportional to pressure, and pressure broadening introduces an additional factor
72 of p , yielding $\tau_{\text{CO}_2} \propto p^2$ (Pierrehumbert 2010; Jeevanjee et al. 2021b). For condensable gases,
73 however, the relationship is less straightforward, because the integrated path does not explicitly
74 depend on pressure. Previous work typically assumes a scaling of the form $\tau \propto p^n$, with n taken
75 empirically to be a larger number such as 4 for water vapor (Frierson et al. 2006).

76 The various approaches described above relate $B_{\tilde{\nu}}$ to τ through coordinate transformations and
77 involve approximations of various levels of empiricism. Here, we seek a more direct mapping

78 from $B_{\bar{\nu}}$ to τ for water vapor. To do so, we draw on prior work formulating the atmospheric
79 vertical structure in the temperature coordinate. Because water vapor acts both as a condensable
80 and a greenhouse gas, expressing the vertical profiles in terms of temperature reveals structural
81 simplicity that is less apparent in traditional vertical coordinates such as height or pressure (Ingram
82 2010; Romps 2014; Jeevanjee 2022; Williams and Jeevanjee 2025).

83 Thermodynamics of dry air and water vapor determine that the temperature lapse rate and relative
84 humidity (Romps 2014) are largely determined by temperature. Specific humidity is given by the
85 product of relative humidity (RH) and saturation specific humidity, and is therefore primarily
86 temperature-controlled (Jeevanjee and Romps 2018). Moreover, the optical thickness from the top
87 of the atmosphere to a given level is approximately proportional to the vertically integrated water
88 vapor path (with modest modifications from pressure broadening), implying that optical depth is
89 likewise largely a function of temperature. Since the Planck function at fixed wavenumber also
90 depends only on temperature, radiative fluxes become, to leading order, functions of temperature
91 alone (Ingram 2010; Jeevanjee and Romps 2018). As a result, increases in surface temperature to
92 leading order extend these relationships to higher temperatures near the surface, without changing
93 the flux profile in the temperature coordinate (Ingram 2010; Jeevanjee and Romps 2018). This
94 leads to the near-invariance of outgoing longwave radiation with surface warming in optically
95 thick regimes, often referred to as Simpson’s law (Simpson 1928; Ingram 2010; Jeevanjee et al.
96 2021a). A related consequence is the fixed-anvil-temperature (FAT) hypothesis (Hartmann and
97 Larson 2002), which links the invariant temperature level at which water vapor can no longer cool
98 efficiently to the detrainment level of anvil clouds.

99 Drawing on previous work on temperature invariance in water vapor thermodynamics and radia-
100 tive transfer, we seek a relation between $B_{\bar{\nu}}$ and τ using temperature as the intermediate variable.
101 When the Planck function is approximated by the Wien approximation and water vapor optical
102 depth by the Clausius-Clapeyron scaling, both $B_{\bar{\nu}}$ and τ exhibit the same exponential dependence
103 on inverse temperature. This leads to a simple power-law relationship between the Planck function
104 and optical depth, with an exponent determined by physical constants and the wavenumber. This
105 power-law formulation appears in the supporting information of Poujol and Bony (2024), where it
106 arises as an intermediate step and is not mentioned in the main text. Here we derive it explicitly and
107 assess its validity in an idealized RCE and line-by-line radiative transfer calculations and assess

the extent to which the approximation holds. We further note that the approximate cancellation of temperature dependence in $B_{\tilde{\nu}}$ and τ is not coincidental, but instead reflects the common appearance of Boltzmann factors, $\exp(-E/k_B T)$, in governing the statistical distribution of microscopic states for both photons and water vapor.

We then explore applications of this power law. First, it enables analytical solutions to the Schwarzschild equations. Second, it provides a spectral theory for column-integrated atmospheric cooling, and familiar results such as Simpson’s law arise naturally. Because this theory does not yield closed-form expressions for cooling, we develop simple algebraic approximations for column-integrated atmospheric cooling and OLR that are convenient for practical use. Finally, we derive a constraint on the cooling-to-space approximation in column-integrated form and show that the smallness of the power-law exponent is key to its accuracy. All of these theoretical results are evaluated against line-by-line radiative transfer calculations.

The paper is organized as follows. Section 2 documents the line-by-line model for testing the theory. Section 3 and 4 are on the $B_{\tilde{\nu}} - \tau$ power law and the solutions to the Schwarzschild equation. Section 5 and 6 are on a spectral theory for column-integrated atmospheric cooling and OLR. Section 7 evaluates the cooling-to-space approximation and discusses its accuracy in the context of the power law. Conclusions and future directions are summarized in Section 8.

2. Line-by-line radiation model and experimental setup

We use the PyRADS radiative transfer model (Koll and Cronin 2018), which incorporates the HITRAN 2016 spectroscopic database (Gordon et al. 2022) and the AER MTCKD continuum model (Mlawer et al. 2012). PyRADS provides spectrally resolved longwave radiative transfer calculations for prescribed temperature and humidity profiles.

We adopt an idealized setup of an atmospheric column that is commonly used in studies focusing primarily on the spectral dimension of radiative transfer rather than its interactions with atmospheric dynamics (e.g. Koll and Cronin 2018; Jeevanjee and Fueglistaler 2020b; Cohen and Pincus 2025). In this setup, the surface radiates as a blackbody at its temperature T_s and the lowest atmospheric level has a temperature equal to T_s . We prescribe either a dry adiabatic or a moist adiabatic lapse rate extending upward from the surface temperature T_s until the temperature reaches 150 K. Above this level, the atmosphere is assumed to be isothermal, representing an idealized stratosphere setting

in at a fixed temperature irrespective T_s (Hartmann and Larson 2002; Seeley et al. 2019). Water vapor is set to zero in the stratosphere for the purposes of this study, as the amount of water vapor in the stratosphere is usually negligible compared to that of the troposphere. Within the troposphere, relative humidity is set to be vertically uniform, with values of 30% or 70%. Surface temperature is varied from 255 to 320 K in 1 K increments.

All calculations are performed at a spectral resolution of 0.01 cm^{-1} . The vertical grid consists of 120 pressure levels evenly spaced in linear pressure. Sensitivity tests using up to 240 levels and linear pressure spacing or logarithmic pressure spacing produce negligible differences in the results.

3. The Planck optical-depth power law

a. Derivation

We layout the theoretical grounds for the expectation that, at wavenumbers dominated by water vapor absorption, the Planck function ($B_{\tilde{\nu}}$) varies with optical depth (τ) approximately as a power law, with a constant exponent at each wavenumber $\tilde{\nu}$.

First, the optical depth scales with temperature approximately following the Clausius–Clapeyron relation. Eq. (2) in the Supporting Information of Koll and Cronin (2018) provides an approximation to the water vapor path in the limit when water vapor makes a negligible contribution to atmospheric pressure and temperature lapse rate, which states that the water vapor path $W = \int \rho q \, dz$ integrated from the tropopause (stratospheric water negligible) to a level within the troposphere of temperature T scales with $T e^*(T)$, where $e^*(T)$ is the saturation vapor pressure given by the Clausius–Clapeyron relation. Eq. (11) in Jeevanjee and Fueglistaler (2020b) uses a similar expression but assumes an arbitrary constant temperature lapse rate and the factor T is replaced by the average temperature between the surface and the tropopause.

Without specifying the exact scaling used for the water vapor path, we simply note that both have the Clausius–Clapeyron scaling which depends on T more strongly than any prefactors, and assume that the optical depth scales with the water vapor path. Although $e^*(T)$ controls the local water vapor concentration, the water vapor path also inherits this scaling. This is because the near-exponential dependence on temperature dominates over both the vertical variation in T and changes in the lapse rate. Integrating an exponential over the temperature range of the troposphere,

166 $\int e^{\alpha T} dT \sim \alpha^{-1} e^{\alpha T}$, approximately preserves the exponential dependence on temperature, provided
 167 that contribution from the lower bound of the integral (tropopause water) is negligible. Therefore,

$$\tau(T) \sim e^{-\frac{L_v}{R_v T}}, \quad (2)$$

168 where L_v is the latent heat of vaporization and R_v is the gas constant for water vapor. The
 169 approximate Clausius-Clapeyron scaling of water vapor optical depth near optically thick band
 170 centers (150 cm^{-1} and 1400 cm^{-1}) and over the optically thin window region (1000 cm^{-1}) are
 171 shown in Figure 1a for four line-by-line calculations with surface temperatures spanning from
 172 275 K to 320 K. Consistent with prior work (Koll and Cronin 2018; Jeevanjee and Romps 2018),
 173 the $\tau(T)$ curve remains approximately invariant and increasing surface temperature only extends
 174 the curve to higher temperatures (Figure 1a).

175 Second, we approximate the Planck function using the Wien approximation,

$$B_{\tilde{\nu}}(T) \approx 2hc^2 \tilde{\nu}^3 \exp\left(-\frac{hc\tilde{\nu}}{k_B T}\right), \quad (3)$$

176 where h is Planck's constant, k_B is Boltzmann's constant, c is the speed of light, and $\tilde{\nu}$ is the
 177 wavenumber. This approximation reproduces the high-wavenumber portion of the spectrum well
 178 and provides a reasonable representation at lower wavenumbers (Figure 1). Following its introduc-
 179 tion in 1896, Wien's formula was quickly superseded by Planck's law in 1900. However, it remains
 180 useful for theoretical work in atmospheric radiation, where analytic tractability is often prioritized
 181 over exact accuracy. Relatively little work in atmospheric radiative transfer has explicitly exploited
 182 the Wien approximation, with Wordsworth et al. (2024) providing a recent example.

190 We may therefore define α and β as the fractional sensitivities of water-vapor optical depth and
 191 the Planck function, respectively, to temperature. From the Clausius-Clapeyron scaling (Eq. 2),
 192 the fractional increase of optical depth is

$$\alpha = \frac{d \ln \tau}{dT} \approx \frac{L_v}{R_v T^2}, \quad (4)$$

193 which has a typical magnitude of approximately $7\% \text{ K}^{-1}$ over the range of temperatures relevant
 194 to the Earth's atmosphere. The corresponding fractional increase of the Planck function under the

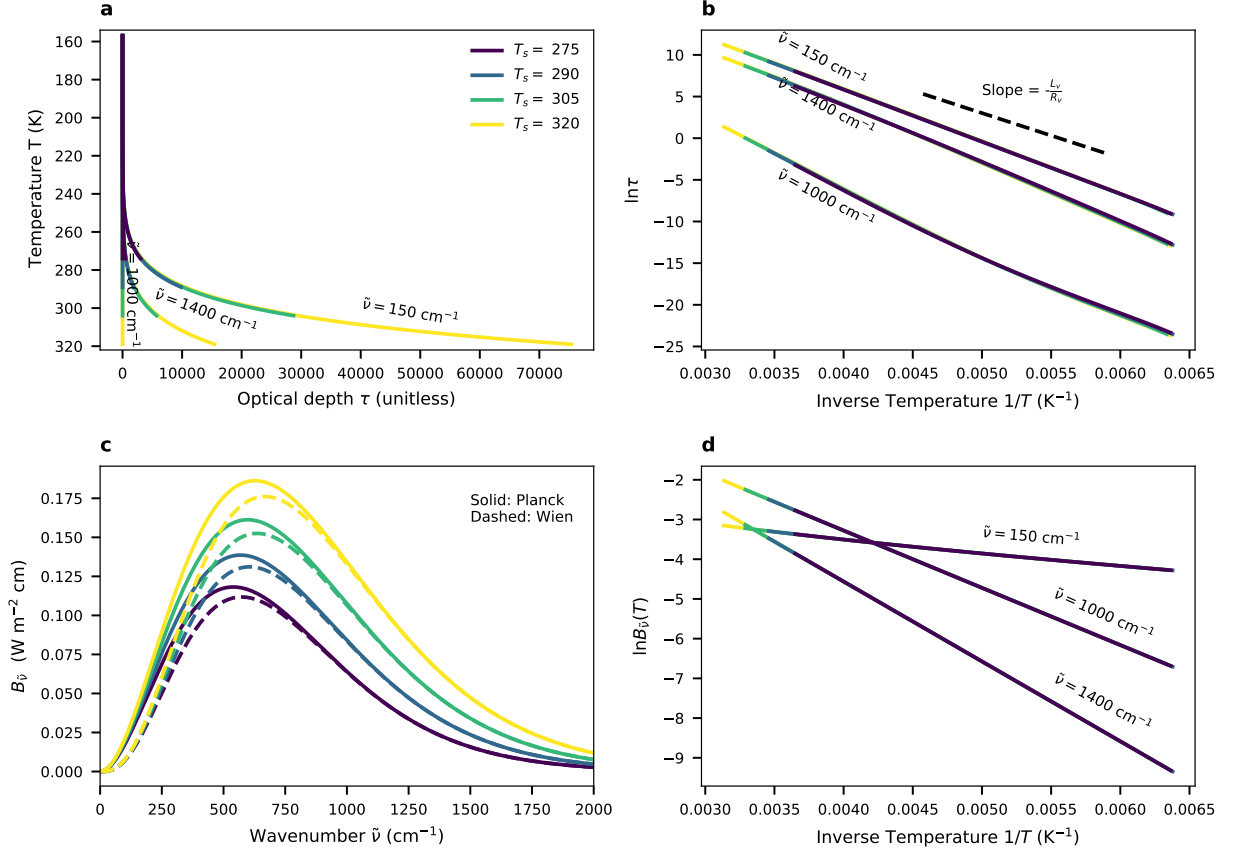


FIG. 1. Optical depth τ and Planck emission $B_{\tilde{\nu}}$ both exhibit an approximately exponential dependence on inverse temperature. (a) Optical depth profiles plotted against temperature show an approximately invariant structure, extending to higher temperatures as T_s increases. Wavenumbers of 150, 1000, and 1400 cm^{-1} are shown. (b) Corresponding near-linear relationships between $\ln \tau$ and $1/T$ for the same wavenumbers and surface temperatures as in (a). (c) The Planck function $B_{\tilde{\nu}}(T)$ (solid) compared with the Wien approximation (dashed) for the same T_s values. (d) Corresponding near-linear relationships between $\ln B_{\tilde{\nu}}(T)$ and $1/T$ for the same wavenumbers and surface temperatures as in (a).

Wien approximation (Eq. 3) is

$$\beta = \frac{d \ln B_{\tilde{\nu}}}{dT} \approx \frac{hc \tilde{\nu}}{k_B T^2}. \quad (5)$$

For a representative terrestrial infrared wavenumber of $\tilde{\nu} \approx 1000 \text{ cm}^{-1}$, β is approximately 2% K^{-1} .

We have emphasized a structural similarity between water-vapor optical depth and Planck emission. This similarity is reflected in the near-linear relationships between $\ln \tau$ and $1/T$ (Figure 1b), and between $\ln B_{\tilde{\nu}}$ and $1/T$ (Figure 1d). The slope of the $\ln \tau$ – $1/T$ relation is similar across the

three wavenumbers and is approximately given by $-L_v/R_v$, whereas the slope of the $\ln B_{\tilde{\nu}}-1/T$ relation is approximately $-hc\tilde{\nu}/k_B$ and varies with wavenumber.

Dividing Eq. (5) by Eq. (4) yields

$$\frac{d \ln B_{\tilde{\nu}}}{d \ln \tau} \equiv \frac{\beta}{\alpha} = \frac{R_v hc \tilde{\nu}}{L_v k_B}. \quad (6)$$

Because both α and β scale as T^{-2} , their ratio, which we define as γ , is independent of temperature, and is, therefore, constant at a given wavenumber $\tilde{\nu}$. As a result, the Planck function at wavenumber $\tilde{\nu}$ can be expressed as a power law in optical depth:

$$B_{\tilde{\nu}}(T(\tau)) \approx B_{\tilde{\nu}}(T_s) \left(\frac{\tau}{\tau_s} \right)^{\gamma}, \quad (7)$$

where the power-law exponent γ is

$$\gamma = \frac{R_v hc \tilde{\nu}}{L_v k_B} \quad (8)$$

and τ_s is the column optical depth or the surface optical depth, and T_s is the surface temperature.

b. Shared Boltzmann factor

Interestingly, the nondimensional parameter γ admits a simple physical interpretation. Using the relation $R_v = k_B/m_v$, where m_v is the molecular mass of water vapor (following from $R_v = R/M_v$ and $k_B = R/N_A$), and defining $l_v = L_v m_v$ as the latent heat released per H₂O molecule, Eq. (8) can be rewritten as

$$\gamma = \frac{h\nu}{l_v} = \frac{\text{photon energy}}{\text{latent heat per H}_2\text{O molecule}}. \quad (9)$$

The non-dimensional parameter γ incorporates two of water vapors roles in the atmosphere, that it condenses and releases latent heat and that it absorbs and emits terrestrial radiation.

A characteristic window-region photon with $\tilde{\nu} = 1000 \text{ cm}^{-1}$ carries an energy of $h\nu = hc\tilde{\nu} \approx 2.0 \times 10^{-20} \text{ J}$, which is about one quarter of the latent heat release associated with condensation of a single water molecule ($l_v \approx 7.5 \times 10^{-20} \text{ J}$). Moving to a different $\tilde{\nu}$, a photon near the maximum of the rotational band ($\tilde{\nu} = 150 \text{ cm}^{-1}$) carries only about 4% of this latent energy l_v . Thus, γ is typically less than one for water vapor in Earth-like climate because a typical terrestrial infrared photon carries only a small fraction of the energy associated with condensation of a single H₂O

221 molecule. We note that the same ratio appears in Eq. (12) Poujol and Bony (2024) as the difference
 222 between emission optical depth and unity; here we attempt to provide a physical interpretation
 223 below.

224 The shared exponential dependence on inverse temperature is instrumental to the power law and
 225 it is not a coincidence. The Wien approximation to the Planck function assumes that the photon
 226 occupation probability follows a Maxwell-Boltzmann distribution of the form $\exp(-E/k_B T)$, where
 227 the relevant energy scale E is the photon energy $h\nu$. The exponential temperature dependence
 228 of water-vapor optical depth follows from the Clausius–Clapeyron relation ($\sim \exp(-L_v/R_v T)$),
 229 which is usually derived from macroscopic thermodynamic laws rather than from a microscopic
 230 energy distribution. However, one may argue that the probability ratio of H₂O molecules in the
 231 gas and liquid phases depends exponentially on the energy difference between these two states,
 232 $\exp(-\Delta E/k_B T)$, if both probabilities are assumed to follow Boltzmann statistics. When the vast
 233 majority of H₂O molecules remain in the liquid phase, the vapor number density and thus the vapor
 234 pressure would then scale with $\exp(-l_v/k_B T)$ (p. 194 of Bohren and Albrecht (2023), Lamb and
 235 Shaw (2016)). In this sense, the shared exponential dependence on inverse temperature ultimately
 236 reflects the generality of the Boltzmann statistics of microscopic states.

237 *c. Validation of the power law with line-by-line model*

249 We first test whether the power-law relation between the Planck function $B_{\tilde{\nu}}$ and optical depth
 250 τ , given by Eq. (7), holds. This relation relies on the assumption that τ scales linearly with
 251 the column-integrated water-vapor path, which is most appropriate for absorption at line centers.
 252 We thus first focus on spectrally isolated lines that clearly exceed neighboring lines in strength
 253 (examples of such lines are given in the insets of Figure 4d). For these cases, the $B_{\tilde{\nu}}-\tau$ power law
 254 holds remarkably well over a wide range of wavenumbers from 200 to 1600 cm⁻¹, as evidenced
 255 by the linear relationship in log–log space (Figure 2a). For the continuum absorption, the log–
 256 log relationship exhibits minimal curvature (Figure A2) and a power law remains an accurate
 257 approximation. As expected from theory, changes in relative humidity do not affect the slope in
 258 log–log space (i.e., the power-law exponent), but instead produce a shift (see Appendix and Figure
 259 A1).

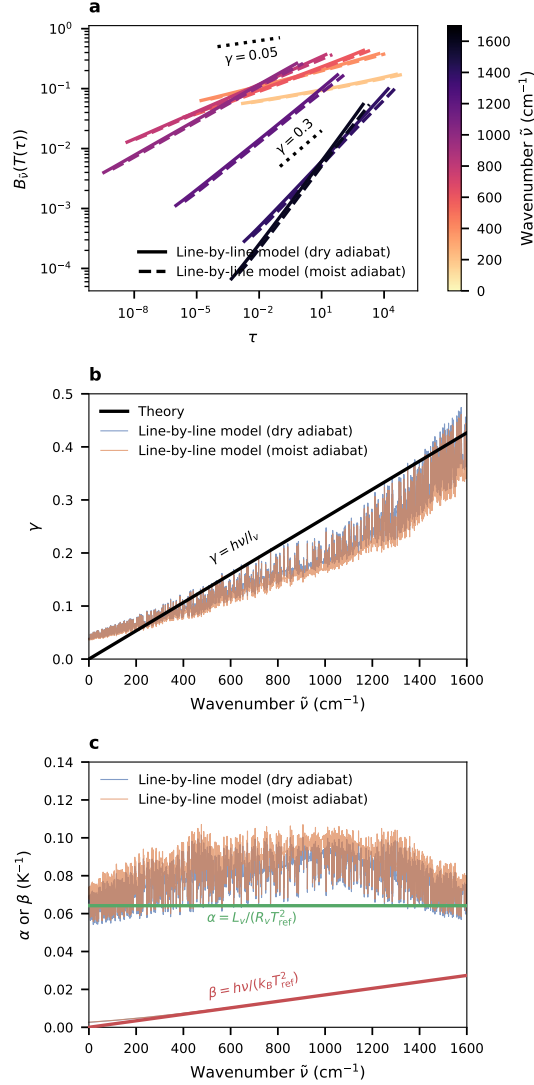


FIG. 2. Approximate power-law relation between the Planck function and optical depth. (a) For a surface temperature $T_s = 290$ K, the Planck function $B_{\tilde{\nu}}(T)$ is shown as a function of τ on logarithmic scales for the closest spectrally isolated lines near the selected wavenumbers (200–1600 cm^{-1} in increments of 200 cm^{-1}). Isolated lines are identified based on τ_s exhibiting a distinct peak and all end up lie within 3 cm^{-1} of the target wavenumbers (see Figure 4d). Only the portion of the profile with temperature greater than 155 K is shown to exclude the stratosphere. Solid and dashed lines correspond to dry and moist adiabatic tropospheres, respectively. (b) Comparison of theoretical and fitted power-law exponents γ across the thermal infrared spectrum. The fitted exponent is obtained from an ordinary least squares (OLS) regression of $\ln B_{\tilde{\nu}}$ against $\ln \tau$, with the slope giving γ (Eq. 8) (c) Comparison of theoretical and fitted values of α and β . The fitted α is the slope obtained from an OLS regression of $\ln \tau$ against $-1/T$ for each wavenumber divided by T_{ref}^2 where $T_{\text{ref}}=290$ K (Eq.4). The fitted β is obtained similarly from an OLS regression of $\ln B_{\tilde{\nu}}$ (Eq.5).

260 We next test whether the theory for γ in Eq. (8) provides a good estimate of the power-law
 261 exponent across the thermal radiation spectrum ($0\text{--}1600\text{ cm}^{-1}$). Comparing theory to the fitted
 262 exponent for each wavenumber from line-by-line calculations of surface temperature 290 K, we
 263 find that the theory captures both the increase of the fitted exponent with wavenumber $\tilde{\nu}$ and its
 264 magnitude (Figure 2b). Spectral lines appear as sharp spikes (both upward and downward) in the
 265 inferred exponent and the continuum absorption appears as the smooth background connecting
 266 these spikes. Agreement with theory is strongest for strongly absorbing wavenumbers, namely the
 267 band center of the vibration–rotation band around 1500 cm^{-1} , and many spectrally isolated lines
 268 at more weakly absorbing wavenumbers (upper tip of spikes in Figure 2b), including those in the
 269 window region. The continuum absorption exhibits a low bias over most wavenumber and the
 270 reason will be discussed in the following paragraph. Using dry-adiabatic versus moist-adiabatic
 271 temperature profiles produces small differences in the inferred exponent with the dry adiabat being
 272 closer to the theory. Overall, the power-law approximation with the exponent given by Eq. (8)
 273 remains broadly valid across the thermal radiation spectrum.

274 Further insight is provided by examining the fractional sensitivities of optical depth and the
 275 Planck function to temperature, namely α (Eq. 4) and β (Eq. 5). α shows the best agreement with
 276 the line-by-line estimate $d \ln \tau / dT$ at strongly absorbing wavenumbers and largely explains both the
 277 agreement and deviations of γ relative to the line-by-line results for wavenumbers above 300 cm^{-1}
 278 (Figure 2a). The deviation of the continuum from the theoretical prediction arises because the actual
 279 optical depth increases more rapidly with temperature than expected from Clausius–Clapeyron
 280 scaling. This is qualitatively consistent with the physical origin of the water vapor continuum,
 281 which is associated with collisions among water vapor molecules. As a result, the optical depth
 282 scales approximately quadratically, rather than linearly, with water vapor mixing ratio (p. 260 of
 283 Pierrehumbert (2010), Koll et al. (2023)), thereby enhancing the temperature sensitivity of $\ln \tau$ (see
 284 Appendix). The deviation of β from the exact derivative $d \ln B_{\tilde{\nu}}(T) / dT$ arises solely from the use
 285 of the Wien approximation (Eq. 3), and is therefore confined to small wavenumbers (Figure 2c),
 286 where it largely accounts for the discrepancy between γ and the line-by-line calculations below
 287 300 cm^{-1} .

4. Approximate solutions to the Schwarzschild equations

The previous section shows that the power law (Eq. 7) provides a simple yet physically grounded approximation for $B_{\tilde{\nu}}(T(\tau))$. This allows the Schwarzschild equation in optical depth space (Eq. 1) to be written in the following form for the upward ($F_{\tilde{\nu}}^{\uparrow}$) and downward ($F_{\tilde{\nu}}^{\downarrow}$) fluxes, with corresponding boundary conditions:

$$\frac{dF_{\tilde{\nu}}^{\uparrow}}{d\tau} - F_{\tilde{\nu}}^{\uparrow} = -\pi B_{\tilde{\nu}}(T_s) \left(\frac{\tau}{\tau_s} \right)^{\gamma}, \quad F_{\tilde{\nu}}^{\uparrow}(\tau_s) = \pi B_{\tilde{\nu}}(T_s), \quad (10)$$

$$\frac{dF_{\tilde{\nu}}^{\downarrow}}{d\tau} + F_{\tilde{\nu}}^{\downarrow} = \pi B_{\tilde{\nu}}(T_s) \left(\frac{\tau}{\tau_s} \right)^{\gamma}, \quad F_{\tilde{\nu}}^{\downarrow}(0) = 0. \quad (11)$$

For the upward flux, the boundary condition is imposed at the surface ($\tau = \tau_s$), where the flux equals the surface Planck emission. For the downward flux, the boundary condition is imposed at the top of the atmosphere ($\tau = 0$), where there is zero incoming longwave radiation. The corresponding solutions to (10) and (11) are

$$F_{\tilde{\nu}}^{\uparrow}(\tau) = \pi B_{\tilde{\nu}}(T_s) \left[e^{-(\tau_s - \tau)} + \int_{\tau}^{\tau_s} \left(\frac{\tau'}{\tau_s} \right)^{\gamma} e^{-(\tau' - \tau)} d\tau' \right], \quad (12)$$

$$F_{\tilde{\nu}}^{\downarrow}(\tau) = \pi B_{\tilde{\nu}}(T_s) \int_0^{\tau} \left(\frac{\tau'}{\tau_s} \right)^{\gamma} e^{-(\tau - \tau')} d\tau', \quad (13)$$

where γ is defined in Eq (8). These solutions provide a good approximation to the line-by-line model results for both dry and moist adiabatic tropospheres, with the dry adiabatic lapse rate obtaining better agreements (Figure 3). Obtaining such approximate analytic solutions may facilitate further theoretical analysis and practical applications.

5. Spectral theory of the column-integrated cooling

a. Derivation

The column-integrated atmospheric longwave cooling at a given wavenumber is the net flux that goes out of the atmospheric column, achieved by computing $F_{\tilde{\nu}}^{\downarrow}(\tau_s) - F_{\tilde{\nu}}^{\uparrow}(\tau_s) + F_{\tilde{\nu}}^{\uparrow}(0)$ from (12)

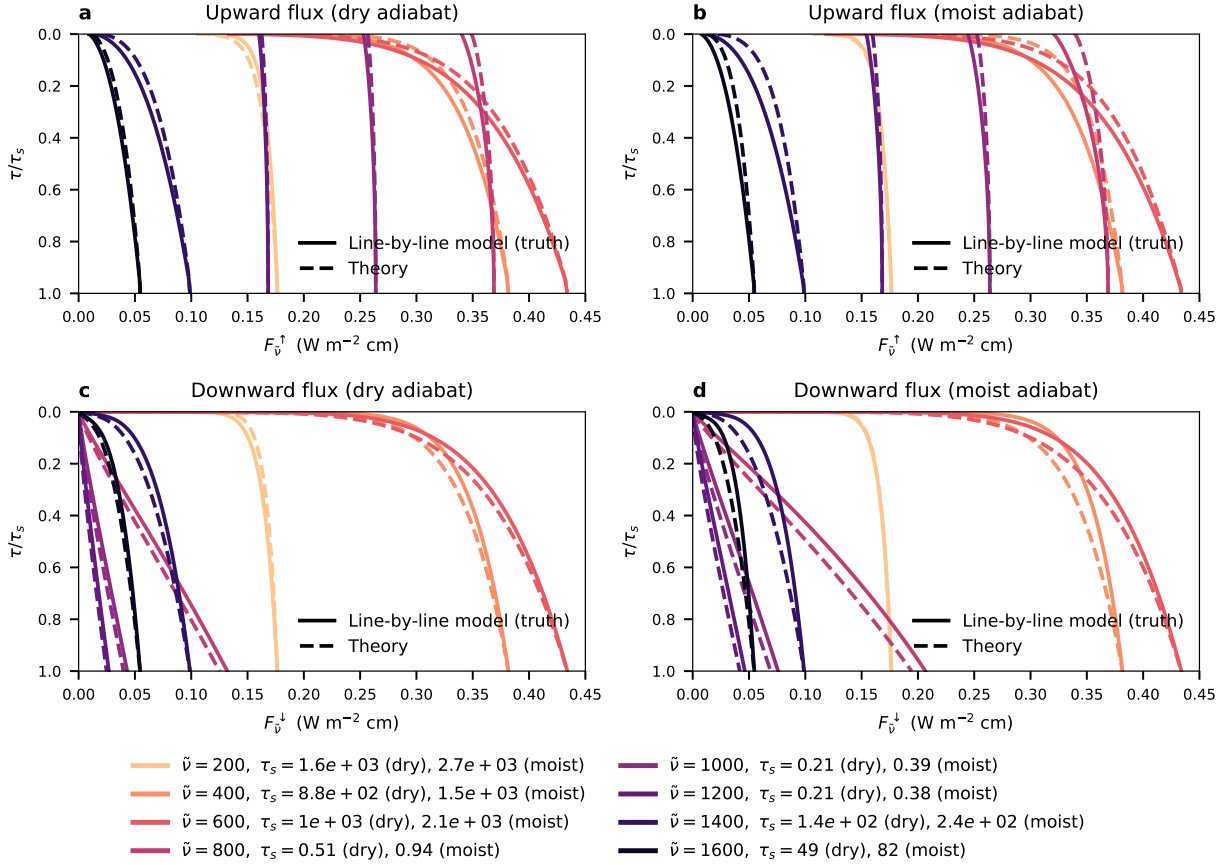


FIG. 3. Comparison of the approximate solutions Eqs. 12 and 13 (dashed) to the Schwarzschild equations with line-by-line model output (solid) for a surface temperature of 290 K. The atmospheric temperature profile follows either a dry adiabat (a, c) or a moist adiabat (b, d). Relative humidity is vertically uniform at 70%. An isothermal stratosphere at 150 K overlies the troposphere and contains no water vapor. Results are shown for wavenumbers 200–1600 cm^{-1} in increments of 200 cm^{-1} and the column optical depths τ_s for each case are listed in the legend.

and (13):

$$\begin{aligned}
 Q_{\tilde{\nu}} = & \underbrace{\pi B_{\tilde{\nu}}(T_s) \int_0^{\tau_s} \left(\frac{\tau}{\tau_s} \right)^{\gamma} e^{-\tau} d\tau}_{\text{cooling to space } (Q_{\tilde{\nu},\text{CTS}})} \\
 & + \underbrace{\pi B_{\tilde{\nu}}(T_s) \int_0^{\tau_s} \left[\left(\frac{\tau}{\tau_s} \right)^{\gamma} - 1 \right] e^{-(\tau_s-\tau)} d\tau}_{\text{exchange with surface } (Q_{\tilde{\nu},\text{EX}})}.
 \end{aligned} \tag{14}$$

where the first term represents radiative exchange between all atmospheric layers and space, whose temperature is effectively 0 K, and is commonly referred to as the cooling-to-space term, while the second term represents radiative exchange between all atmospheric levels and the surface.

The cooling-to-space approximation (Green 1967; Jeevanjee and Fueglistaler 2020a) is a commonly invoked concept in atmospheric radiation. In its column-integrated form, this approximation states that the exchange-with-surface term is negligible, such that atmospheric cooling can be approximated by the cooling-to-space contribution alone,

$$Q_{\tilde{\nu}} \approx Q_{\tilde{\nu},\text{CTS}}. \quad (15)$$

The accuracy of this approximation is analyzed in Section 7. For now, we have

$$Q_{\tilde{\nu}} \approx \pi B_{\tilde{\nu}}(T_s) \Gamma(1 + \gamma, \tau_s) \tau_s^{-\gamma}, \quad (16)$$

where $\Gamma(1 + \gamma, \tau_s)$ denotes the lower incomplete gamma function, $\Gamma(1 + \gamma, \tau_s) \equiv \int_0^{\tau_s} \tau^\gamma e^{-\tau} d\tau$. The use of the incomplete Gamma function in atmospheric radiative transfer has precedents in Pierrehumbert (2010); Robinson and Catling (2012); Jeevanjee et al. (2021b); Tang and McColl (2025), as this form arises when a power-law relationship between the Planck emission $B_{\tilde{\nu}}(T)$ and optical depth τ is assumed. The main distinction here is that the exponent is determined by physical constants for each wavenumber (Eq. 8), whereas in prior work the power law is primarily a mathematical convenience and the exponent may vary with τ .

b. Thick limit

We now examine the limiting behavior of Eq. (16). For very large column optical depth τ_s , corresponding to strongly absorbing water–vapor bands, the upper limit of the integral in Eq. (16) may be taken to infinity, since the integrand decays exponentially with τ . In this limit, the incomplete gamma function reduces to the ordinary gamma function, yielding

$$Q_{\tilde{\nu}} \approx \pi B_{\tilde{\nu}}(T_s) \Gamma(1 + \gamma) \tau_s^{-\gamma}, \quad \tau_s \gg 1. \quad (17)$$

Because $\Gamma(1 + \gamma)$ deviates from unity by at most $\sim 10\%$ for $0 \leq \gamma \leq 1$, Eq. (17) admits an even simpler thick-limit form,

$$Q_{\tilde{\nu}} \approx \pi B_{\tilde{\nu}}(T_s) \tau_s^{-\gamma}, \quad \tau_s \gg 1. \quad (18)$$

The key implication is that, for optically thick wavenumbers, both atmospheric cooling and outgoing longwave radiation (OLR) follow a power law with exponent $-\gamma$ (OLR is well approximated by $Q_{\tilde{\nu},\text{CTS}}$ in this regime.) We make three remarks related to this.

(i) *Weak sensitivity of cooling to RH.* Because γ is small for wavenumbers relevant to thermal infrared radiation, optically thick atmospheric cooling depends only weakly on the column optical depth τ_s . For example, taking $\gamma = 0.15$, representative of strong water–vapor cooling near $\tilde{\nu} \approx 600 \text{ cm}^{-1}$ (Figure 5a), a tenfold increase in τ_s reduces the cooling to about 70%, a thousandfold increase to roughly 30%, and even a millionfold increase only to 10%. Thus, very large changes in optical depth produce only modest changes in atmospheric cooling at fixed temperature.

This weak sensitivity has direct implications for relative humidity. If surface and atmospheric temperatures are held fixed, variations in relative humidity (RH) modify cooling through changes in τ_s , yielding

$$\frac{\partial \ln Q_{\tilde{\nu}}}{\partial \ln \text{RH}} = -\gamma. \quad (19)$$

Hence, column-integrated cooling depends on RH through a power law with a small negative exponent. For typical free-tropospheric variations from 30% to 90% RH (a threefold increase in τ_s), this scaling implies a reduction in cooling of about 16% near $\tilde{\nu} \approx 600 \text{ cm}^{-1}$ and 34% near $\tilde{\nu} \approx 1400 \text{ cm}^{-1}$, at fixed surface temperature. This result is broadly consistent with the finding that tropospheric longwave cooling varies only weakly across climate zones with differing free-tropospheric RH (e.g., Figure 1 of Jeevanjee and Fueglistaler (2020b)), although previous arguments are regarding spectrally integrated, vertically resolved cooling.

(ii) *The “ $\tau = 1$ ” law.* Equation (18) may also be interpreted as Planck emission originating from the level $\tau = 1$, once the power-law relation between the Planck function and optical depth in Eq. (7) is assumed:

$$Q_{\tilde{\nu}} \approx \pi B_{\tilde{\nu}}(T(1)). \quad (20)$$

This recovers the familiar “ $\tau = 1$ ” law.

Retaining the $\Gamma(1 + \gamma)$ factor instead of approximating it as unity, Eq. (17) may be rewritten as

$$Q_{\bar{\nu}} \approx \pi B_{\bar{\nu}}(T(\tau_{\text{eff}})), \quad \tau_{\text{eff}} \equiv \Gamma(1 + \gamma)^{\frac{1}{\gamma}}. \quad (21)$$

The same formulation appears in Appendix B of (Jeevanjee et al. 2021b), where γ is defined more generally and estimated for CO_2 and water vapor. Using the present theory for γ , we obtain consistent values: at 600 cm^{-1} , $\tau_{\text{eff}} \approx 0.63$ for $\gamma = 0.15$, and at 1500 cm^{-1} , $\tau_{\text{eff}} \approx 0.74$ for $\gamma = 0.4$. These results confirm that the effective emission level remains of order unity for water vapor (Huang and Bani Shahabadi 2014; Jeevanjee and Romps 2018; Jeevanjee and Fueglistaler 2020b).

(iii) *Simpson's law.* Simpson's law states that in an atmosphere with constant relative humidity, the OLR at water-vapor-dominated wavenumbers does not change with surface warming (Simpson 1928; Ingram 2010; Jeevanjee et al. 2021a). Taking the derivative of the thick limit in Eq. (17) with respect to surface temperature T_s gives

$$\frac{dQ_{\bar{\nu}}}{dT_s} \approx \frac{B_{\bar{\nu}}(T_s)}{\tau_s^\gamma} \left(\frac{d \ln B_{\bar{\nu}}(T_s)}{dT_s} - \gamma \frac{d \ln \tau_s}{dT_s} \right) = 0. \quad (22)$$

The last step follows from the definitions of α , β , and γ (Eqs. 4, 5, and 8). The power-law relation between the Planck function and optical depth thus automatically satisfies the invariance of OLR with respect to surface temperature changes and represents a simple analytical formulation of Simpson's law.

c. *Thin limit*

For wavenumbers where water vapor is not strongly absorbing and τ of any atmospheric level is much less than one, the exponential term ($e^{-\tau}$) in the integrand of Eq. (16) can be approximated as unity, yielding

$$Q_{\bar{\nu}} \approx \pi B_{\bar{\nu}}(T_s) \frac{\tau_s}{1 + \gamma}, \quad \tau_s \ll 1. \quad (23)$$

In this limit, atmospheric cooling scales linearly with the column optical depth.

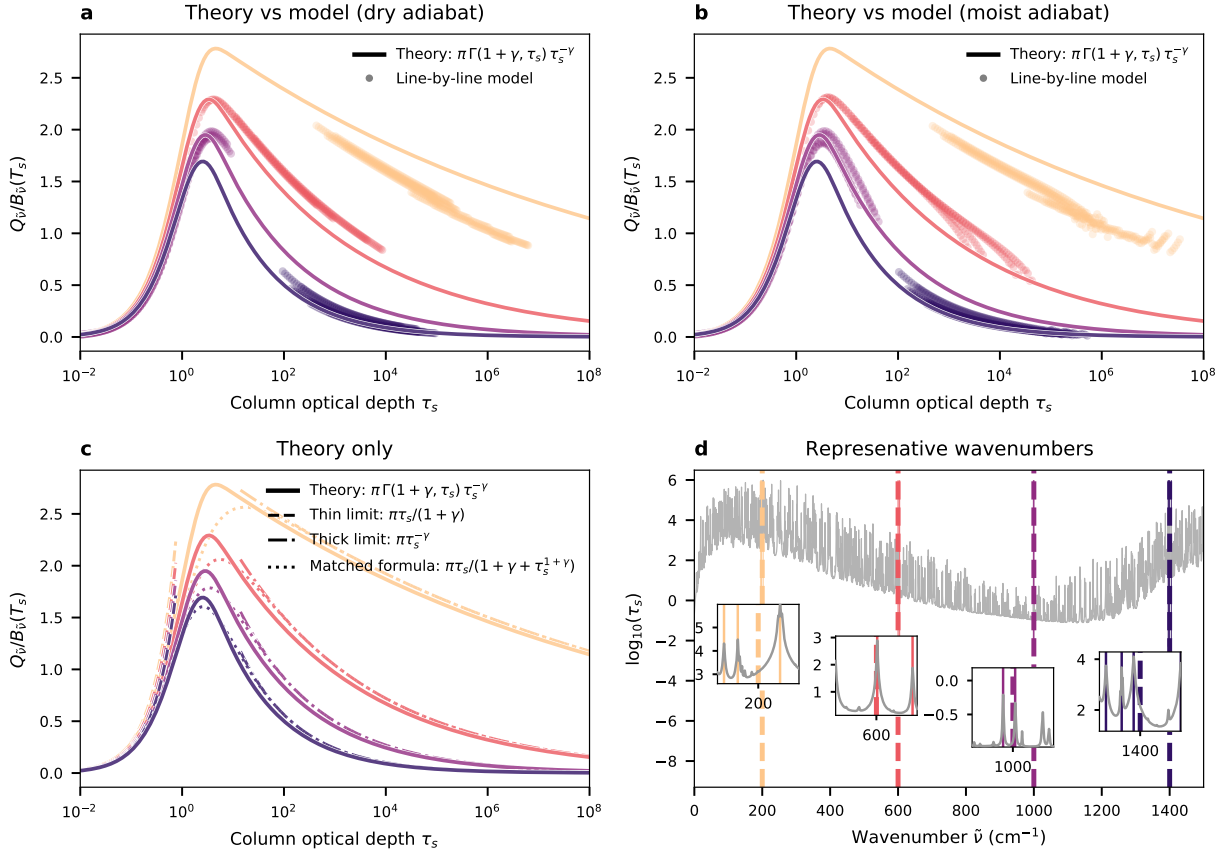


FIG. 4. Theory for atmospheric cooling (Eq. ??), its limits, and comparison with line-by-line calculations. (a) Column-integrated atmospheric cooling, normalized by the surface Planck emission, $Q_{\tilde{\nu}}/B_{\tilde{\nu}}(T_s)$, as a function of column optical depth τ_s for representative wavenumbers 200, 600, 1000, and 1400 cm^{-1} . Solid curves show the theory (Eq. 16). Dots indicate line-by-line radiative transfer calculations from PyRADs for water-vapor absorption lines within $\pm 5 \text{ cm}^{-1}$ of each representative wavenumber. For each line, 130 data points are shown, corresponding to experiments with surface temperatures ranging from 255 K to 320 K, dry adiabatic lapse rates in the troposphere, and vertically uniform relative humidities of 30% and 70%. (b) Same as (a), but for experiments with moist adiabatic lapse rates in the troposphere. (c) The incomplete-Gamma-function theory, thin and thick limits (Eq. 24), and the formula matching both limits (Eq. 25) for the same wavenumbers. (d) Logarithm of column optical depth, $\log_{10}(\tau_s)$, as a function of wavenumber $\tilde{\nu}$ with zoomed-in insets for a reference state ($T_s = 290 \text{ K}$, $\text{RH} = 0.7$) with a dry adiabatic lapse rate. Vertical colored lines mark the spectral lines shown in panels (a) and (b).

376 *d. Validation of cooling theory with line-by-line model*

389 We test the lower incomplete Gamma-function approximation for column-integrated cooling $Q_{\tilde{\nu}}$
390 (Eq. 16) against line-by-line calculations. Because the exponent γ varies with wavenumber $\tilde{\nu}$
391 (Eq. 8), the dependence of $Q_{\tilde{\nu}}$ on column optical depth τ_s is likewise wavenumber dependent. We
392 therefore evaluate the theory for water-vapor absorption lines near representative wavenumbers
393 200, 600, 1000, and 1400 cm^{-1} (Figure 4d), which together span much of the thermal infrared
394 spectrum and cover both the optically thin and thick regimes (Figure 4a,b).

395 In Eq. (16), the explicit dependence of $Q_{\tilde{\nu}}$ on surface temperature enters only through the
396 prefactor $B_{\tilde{\nu}}(T_s)$. After normalization by $B_{\tilde{\nu}}(T_s)$, the theory predicts that results from different
397 surface temperatures collapse onto a single curve when plotted against τ_s . This expectation is
398 largely borne out by the line-by-line calculations: data spanning surface temperatures from 255 K
399 to 320 K fall rather close to the theoretical curves. The normalized cooling increases linearly with
400 τ_s in the optically thin regime, reaches a maximum for $\tau_s \sim \mathcal{O}(1)$, and decreases as a power law
401 with exponent $-\gamma$ in the optically thick limit (Figure 4a, b, and c).

402 The theory (Eq. 16) exhibits a high bias for $\tilde{\nu} \approx 200 \text{ cm}^{-1}$ and smaller wavenumbers (not shown),
403 but performs better at larger wavenumbers. This bias is likely due to the breakdown of the Wien
404 approximation at low wavenumbers. As shown in Figure 2c, the Wien-based estimate of β is
405 smaller than the Planck-based estimate, which propagates into an underestimate of the exponent γ .
406 In the optically thick limit, where $Q_{\tilde{\nu}} \sim \tau_s^{-\gamma}$, this leads to a decay that is too slow with increasing
407 τ_s . The actual decay is therefore steeper, resulting in an overestimate of $Q_{\tilde{\nu}}$ at small wavenumbers.

408 The line-by-line calculations also confirm the lack of explicit dependence of Eq. (16) on RH. Data
409 from experiments with RH of 30% and 70% form nearly indistinguishable curves (Figure 4a,b),
410 consistent with the theory, in which RH enters only implicitly through its effect on the column
411 optical depth τ_s .

412 **6. Matched formulae for spectral cooling and OLR**

413 Although the lower incomplete Gamma function is analytical and based on physical assumptions,
414 it does not admit a closed-form expression and may therefore be less convenient for practical use.
415 We thus provide simpler matched formulae that capture the asymptotic behavior of the incomplete

416 gamma function, ensuring that the resulting approximation to the atmospheric cooling spectrum
 417 remains reasonably accurate while being easier to use.

418 The incomplete-Gamma-function formulation of atmospheric cooling exhibits the following
 419 asymptotic behavior:

$$Q_{\tilde{\nu}} \sim \pi B_{\tilde{\nu}}(T_s) \begin{cases} \frac{\tau_s}{1+\gamma}, & \tau_s \rightarrow 0, \\ \tau_s^{-\gamma}, & \tau_s \rightarrow \infty, \end{cases} \quad (24)$$

420 where γ is the same exponent defined in Eq. (8). Matching the two limits in Eq. (24) yields the
 421 following expression:

$$Q_{\tilde{\nu}} \approx \pi B_{\tilde{\nu}}(T_s) \frac{\tau_s}{1 + \gamma + \tau_s^{1+\gamma}}. \quad (25)$$

422 This expression transitions smoothly between the optically thin and thick regimes and provides a
 423 good approximation to the full lower incomplete Gamma-function expression (Eq. 16) across the
 424 range of γ relevant for the thermal spectrum (Figure 4c), while retaining the advantage of being
 425 fully algebraic.

426 Taking as input the surface temperature T_s and the column optical depth τ_s of each wavenumber
 427 $\tilde{\nu}$ we find that the theory (Eq. 16) is in good agreement with the spectrum of column-integrated
 428 atmospheric cooling (Figure 5a). Moderate underestimates near 600 cm^{-1} exist for the moist
 429 adiabatic atmosphere. The matched formula in Eq. (25) also achieves a similar level of agreement
 430 over most of the spectrum.

431 We are able to treat the OLR spectrum similarly. The OLR at the top of the atmosphere is obtained
 432 by evaluating the upward flux (Eq. 12) at $\tau = 0$. This consists of the sum of the atmospheric cooling-
 433 to-space term (first term in Eq. 14) and the attenuated surface emission. Replacing $Q_{\tilde{\nu}, \text{CTS}}$ with the
 434 incomplete Gamma-function expression (Eq. 16) yields

$$F_{\tilde{\nu}}^{\uparrow}(0) = \underbrace{\pi B_{\tilde{\nu}}(T_s) \Gamma(1+\gamma) \tau_s^{-\gamma}}_{\text{atmospheric emission } (=Q_{\tilde{\nu}, \text{CTS}} \approx Q_{\tilde{\nu}})} + \underbrace{\pi B_{\tilde{\nu}}(T_s) e^{-\tau_s}}_{\text{surface emission}}. \quad (26)$$

435 In the optically thick limit ($\tau_s \gg 1$), the surface contribution is exponentially suppressed, and
 436 the OLR approaches the atmospheric cooling-to-space term. In the optically thin limit ($\tau_s \ll 1$),
 437 atmospheric emission becomes negligible and the OLR approaches the surface Planck emission,

438 $\pi B_{\tilde{\nu}}(T_s)$. The asymptotic behavior of $F_{\tilde{\nu}}^{\uparrow}(0)$ is therefore

$$F_{\tilde{\nu}}^{\uparrow}(0) \sim \begin{cases} \pi B_{\tilde{\nu}}(T_s), & \tau_s \rightarrow 0, \\ \pi B_{\tilde{\nu}}(T_s) \tau_s^{-\gamma}, & \tau_s \rightarrow \infty, \end{cases} \quad (27)$$

439 where γ is defined in Eq. (8).

440 A simple matched expression connecting these limits is

$$F_{\tilde{\nu}}^{\uparrow}(0) \approx \frac{\pi B_{\tilde{\nu}}(T_s)}{(1 + \tau_s)^{\gamma}}, \quad (28)$$

441 which provides a compact analytic representation of the OLR spectrum as a function of optical
442 depth and wavenumber. A different spectral theory for OLR has been proposed in Appendix C of
443 Jeevanjee and Fueglistaler (2020a).

444 Using T_s and τ_s as input, we find that Eq. (26 and the matched formula Eq. (28) both reproduce
445 the OLR spectrum for a reference temperature of 290 K with high fidelity (Figure 5b).

452 7. Accuracy of column-integrated cooling-to-space approximation

462 The theory of the column-integrated cooling spectrum in Section 5 is based on the cooling-
463 to-space approximation, which neglects the exchange-with-surface term ($Q_{\tilde{\nu},\text{EX}}$) in Eq. (14). The
464 approximation is accurate across most of the spectrum, as illustrated in Fig. 6a, but here we formally
465 estimate the accuracy of this approximation by estimating $Q_{\tilde{\nu},\text{EX}}$.

466 Taylor expanding the Planck function $B_{\tilde{\nu}}(T(\tau))$ about τ_s and retaining terms up to linear order
467 yields

$$Q_{\text{EX}} \approx -\pi \left. \frac{dB_{\tilde{\nu}}}{d\tau} \right|_{\tau=\tau_s} \int_0^{\tau_s} (\tau_s - \tau) e^{-(\tau_s - \tau)} d\tau \quad (29)$$

$$= -\pi \gamma \frac{B_{\tilde{\nu}}(T_s)}{\tau_s} [1 - (\tau_s + 1)e^{-\tau_s}], \quad (30)$$

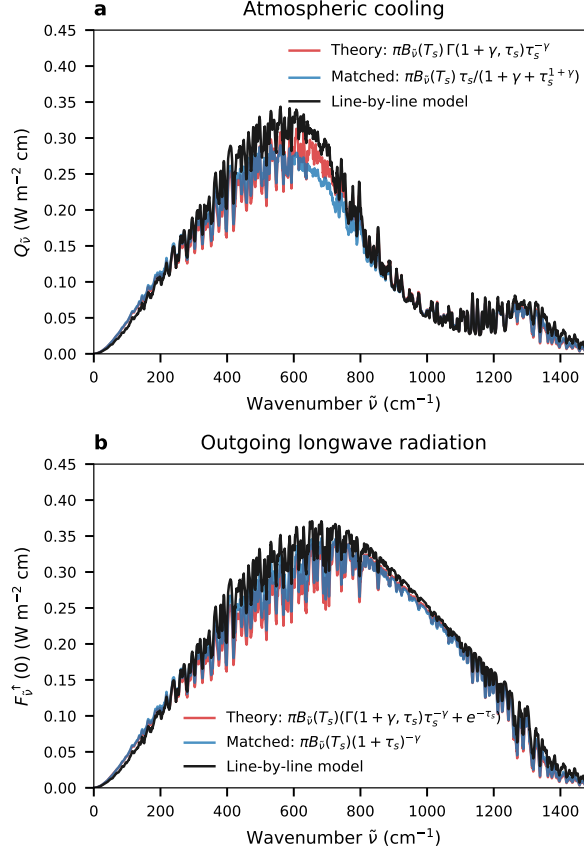


FIG. 5. Spectra of column-integrated cooling and outgoing longwave radiation compared with various approximations. Line-by-line results for a column with surface temperature 290 K, relative humidity 70%, and a dry adiabatic troposphere are shown in black in both panels. (a) Column-integrated cooling $Q_{\tilde{\nu}}$: the matched formula (Eq. 25) is shown in blue, and the theory (Eq. 16) is shown in red. (b) Outgoing longwave radiation: the matched formula (Eq. 28) is shown in blue, and the corresponding theory (Eq. 26) is shown in red. All spectra are smoothed using a rolling average over 5 cm⁻¹.

where the second line follows from evaluating $dB_{\tilde{\nu}}/d\tau$ at τ_s using the Wien approximation and Clausius–Clapeyron scaling (Eqs. 4 and 5), together with $\gamma = \beta/\alpha$, giving

$$\left. \frac{dB_{\tilde{\nu}}}{d\tau} \right|_{\tau=\tau_s} = \gamma \frac{B_{\tilde{\nu}}(T_s)}{\tau_s}. \quad (31)$$

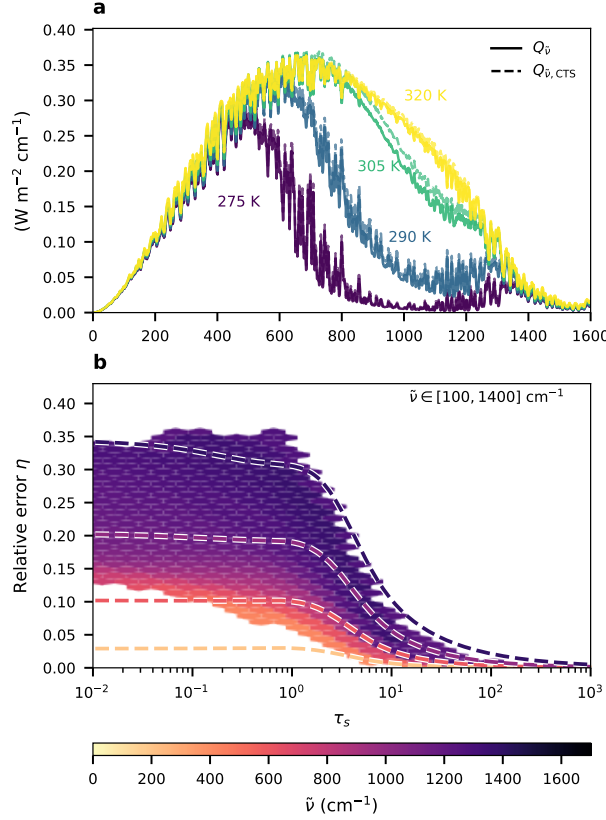


FIG. 6. The accuracy of the column-integrated cooling-to-space approximation is controlled by the smallness of γ . (a) Spectra of column-integrated atmospheric longwave cooling (solid) and the cooling-to-space term alone (dashed), computed from line-by-line radiative-transfer calculations for dry-adiabatic temperature profiles. See Eq. (14) for definition of the terms. Results are shown for surface temperatures ranging from 275 to 320 K in 15 K increments and vertically constant relative humidity of 70%. (b) Relative error η of the column-integrated cooling-to-space approximation. Hexagons show line-by-line results for surface temperatures ranging from 255 K to 320 K, wavenumbers from 100 to 1400 cm^{-1} , and RH values 0.3 and 0.7 binned in $(\tau_s, \tilde{\nu})$ space and colored by the median $\tilde{\nu}$. Only bins with more than 50 data points are shown (there is a total of 17 million data points). Superposed theoretical curves (Eq. 33) correspond to wavenumbers 200, 600, 1000 and 1400 cm^{-1} .

The relative error incurred by neglecting $Q_{\tilde{\nu}, \text{EX}}$ is

$$\eta \equiv -\frac{Q_{\tilde{\nu}, \text{EX}}}{Q_{\tilde{\nu}, \text{CTS}} + Q_{\tilde{\nu}, \text{EX}}}. \quad (32)$$

Substituting $Q_{\tilde{\nu},\text{CTS}}$ with the matched expression Eq. (25) and $Q_{\tilde{\nu},\text{EX}}$ with Eq. (30), η then depends only on τ_s for each $\tilde{\nu}$ (dashed lines in Fig. 6b).

As $\tau_s \rightarrow \infty$, $\eta \rightarrow 0$, and the maximum error occurs in the optically thin limit $\tau_s \rightarrow 0$, for which

$$\lim_{\tau_s \rightarrow 0} \eta = \frac{\gamma(1 + \gamma)}{2 - \gamma(1 + \gamma)} \quad (33)$$

With the approximation for γ (Eq. 8), we estimate the maximum η to range from 0.02% for 150 cm⁻¹ near the rotational band center to 34% for 1400 cm⁻¹ near the right tail of Earth’s Planck emission. At wavenumbers corresponding to the strongest cooling (approximately 400–800 cm⁻¹), the maximum η is modest, ranging from about 6% to 15%.

Eq. (33) suggests that the smaller γ is, the smaller the relative error induced by the column-integrated cooling-to-space approximation. This is qualitatively consistent with Jeevanjee and Fueglistaler (2020a), who argue that $\gamma \equiv d \ln B_{\tilde{\nu}} / d \ln \tau \ll 1$ is the criterion for the cooling-to-space approximation to hold near $\tau = 1$ when $\tau_s \gg 1$, whereas our argument concerns the column-integrated cooling-to-space approximation and is not limited to $\tau_s \gg 1$.

Combined with the interpretation of γ for a condensable gas as the ratio of photon energy to the latent heat released per molecule (Eq. 9), this implies that the cooling-to-space approximation is accurate in Earth-like climates but may not hold for RCE involving other condensable gases, depending on the latent heat of the gas and the wavenumbers with which it interacts.

8. Conclusions and future directions

We show that, over water–vapor–dominated wavenumbers, the Planck function $B_{\tilde{\nu}}$ varies approximately as a power law in optical depth τ (Eq. 7). The exponent γ admits a simple expression as fundamental physical constants and wavenumber (Eq. 8) and is given by the ratio of the energy of a terrestrial infrared photon to the latent heat released per water molecule (Eq. 9). This parameter γ encapsulates the dual physical role of water vapor as a condensable constituent of the atmosphere and as a radiatively active greenhouse gas. Because the latent heat of vaporization is large compared to the energy of a terrestrial infrared photon, γ remains small across the thermal infrared spectrum (Figure 2). The convenience of the Wien approximation to the Planck function demonstrated here might be useful in future work as well. We further argue that the power law arises from the shared

Boltzmann factor governing the energy distributions of water vapor molecules and photons (in the classical limit).

Using this Planck–optical-depth power law, we provide approximate solutions to the Schwartzchild equations for upward and downward longwave radiative fluxes (Eq. 12 and 13). We also derive an expression for the column-integrated cooling-to-space spectrum (Eq. 16). This expression unifies the optically thin and optically thick limits (Figure 4a-c) and reproduces the atmospheric cooling spectrum from line-by-line calculations (Figure 5).

The atmospheric cooling theory predicts limiting behaviors. In the optically thin limit ($\tau_s \rightarrow 0$), atmospheric cooling scales linearly with the column optical depth τ_s , while in the optically thick limit ($\tau_s \rightarrow \infty$) it follows a power law in τ_s with exponent $-\gamma$. Because γ is small, This scaling implies that atmospheric cooling decreases only weakly with increasing optical depth. This is equivalent to holding the surface temperature constant while increasing the relative humidity and consistent with that atmospheric cooling is insensitive to relative humidity. The same power law also yields a simple analytical formulation of Simpson’s law by automatically satisfying the invariance of OLR to surface warming (Eq. 22).

For ease of future application, we develop a closed-form expression for atmospheric cooling (Eq. 25) that reproduce the optically thin and optically thick limits of the incomplete Gamma-function theory. We also derive a corresponding matched expression for the OLR spectrum (Eq. 28). Comparisons with line-by-line radiative-transfer calculations demonstrate that these matched formulae accurately capture the spectral structure of both atmospheric cooling and OLR (Figure 5).

Lastly, we analyze the accuracy of the cooling-to-space approximation and find that the accuracy for the column-integrated cooling is ensured by the smallness of the nondimensional parameter γ . The smallness of γ traces back to the large latent heat of a water vapor molecule compared to the photon energy of thermal radiation.

This work can be extended in several directions. The present theory has been developed and tested for idealized column atmospheres with vertically uniform relative humidity and adiabatic temperature profiles. A natural next step is to evaluate its performance using observed atmospheric temperature and humidity profiles. Local and transient phenomena that deviate from column-mean behavior, such as temperature inversions, may become important and challenge the theory.

527 The spectral theories depending on the column optical depth and surface temperature presented
 528 here could be combined with simple spectral models (Jeevanjee and Fueglistaler 2020b) to yield
 529 spectrally-integrated theories. Finally, applying similar theories to other condensable greenhouse
 530 gases or to non-Earth atmospheres may be possible.

531 *Acknowledgments.* The author thanks Daniel Koll for making the PyRADS model publicly avail-
 532 able, which makes use of HITRAN 2016 line lists (<http://hitran.org/>) and AER's MTCKD
 533 continuum model (http://rtweb.aer.com/continuum_frame.html). The author thanks Sean
 534 Cohen, Nadir Jeevanjee, Robert Pincus, Adam Sobel, and Brett McKim for helpful feedback.

535 *Data availability statement.* All data used in the paper are model output of the PyRADS model
 536 which is freely available at <https://github.com/danielkoll/PyRADS>.

537 APPENDIX

538 **Effect of relative humidity and continuum absorption on the power law**

539 The relative humidity (RH) does not affect the power-law scaling between $B_{\tilde{\nu}}$ and τ as long as
 540 it is vertically uniform. In that case, RH simply rescales the column optical depth τ_s , leading to
 541 a vertical shift in log–log space without changing the slope. Assume two profiles with identical
 542 temperature structure but different constant RH values r_1 and r_2 and column optical depth $\tau_{s,1}$ and
 543 $\tau_{s,2}$. Since optical depth is proportional to the water-vapor path, we have

$$\frac{\tau_{s,1}}{\tau_{s,2}} = \frac{r_1}{r_2}. \quad (\text{A1})$$

544 From the power-law relation in Eq. (7),

$$\ln B_{\tilde{\nu},1} = \gamma \ln \tau - \gamma \ln \tau_{s,1} \quad (\text{A2})$$

$$\ln B_{\tilde{\nu},2} = \gamma \ln \tau - \gamma \ln \tau_{s,1} \quad (\text{A3})$$

$$(\text{A4})$$

545 The two curves should then be offset by $\gamma \ln(r_1/r_2)$ in the log–log plot while the slope remains
 546 unchanged, as shown in Figure A1.

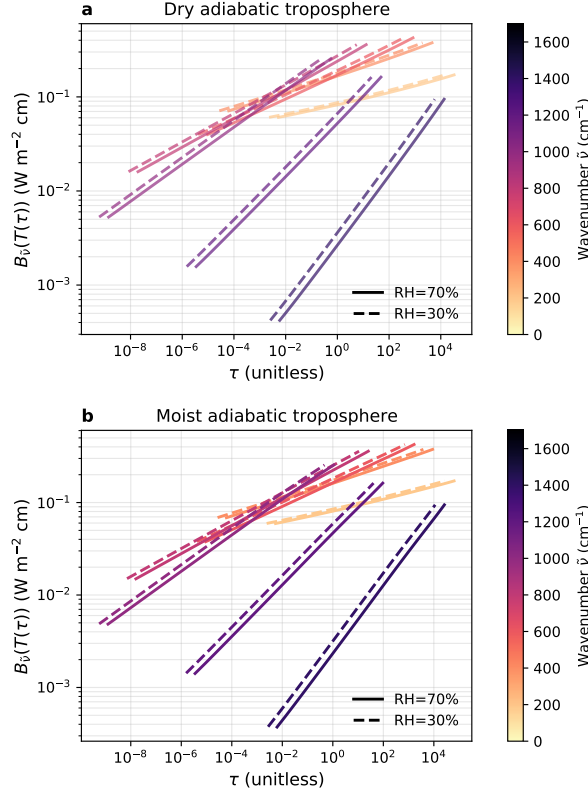


FIG. A1. Effect of relative humidity on the power-law scaling as a vertical shift in log–log space. (a) Dry adiabatic troposphere with surface temperature 290 K and vertically uniform relative humidity of 70% (solid) and 30% (dashed). (b) Same as (a), but for a moist adiabatic troposphere. Wavenumbers span 200–1600 cm^{-1} in increments of 200 cm^{-1} . The standalone lines, as in Figure 2a, are selected based on τ_s exhibiting a distinct peak of the target wavenumbers.

The water vapor continuum is dominated by self-collisions and, as a result, the optical depth scales approximately quadratically, rather than linearly, with water vapor mixing ratio (p. 260 of Pierrehumbert (2010); Koll et al. (2023)). Therefore,

$$\tau_{\text{cnt}} \sim e^{-\frac{2L_v}{R_v T}}, \quad (\text{A5})$$

where the subscript “cnt” denotes the continuum. This suggests that

$$\frac{d \ln \tau_{\text{cnt}}}{dT} \approx \frac{2L_v}{R_v T^2}. \quad (\text{A6})$$

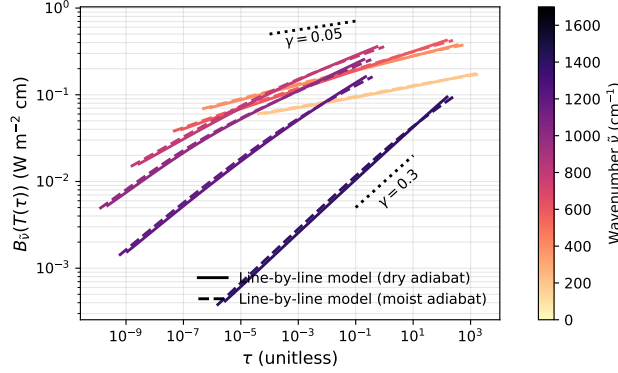


FIG. A2. Relationship between $B_{\tilde{\nu}}$ and τ , as in Figure 2a, but evaluated at wavenumbers exactly from 200 to 1600 cm^{-1} in increments of 200 cm^{-1} , rather than at the closest isolated lines.

Thus, for the same β , the exponent for a continuum is

$$\gamma_{\text{cnt}} = \frac{1}{2} \gamma, \quad (\text{A7})$$

where γ is defined in Eq. (8). This is smaller than the fitted exponent for the continuum in Figure 2b, but is consistent with the tendency toward reduced γ .

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